Domain Adapted AI Assistant

Team Members

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Domain Task Definition

The goal of this project was to improve our GenAI system so it can give better answers for computer science questions, mainly related to operating systems. Before this, the system would sometimes give answers that were too general or not really focused on the topic.

So the idea was to make it more like a domain-specific assistant instead of just a regular chatbot. We wanted it to explain concepts in a way that actually makes sense for someone learning the material.

Instruction Dataset Description

We created an instruction dataset with question and answer pairs based on operating systems topics. Each example includes a question and a correct type of answer that the model should learn from.

At the beginning, the dataset was really small, so the responses were kind of inconsistent and sometimes off-topic. After adding more examples, the model started giving better and more stable answers. It wasn't perfect, but there was definitely an improvement.

Adaptation Method Used

We used LoRA (Low-Rank Adaptation) with the PEFT library. Instead of retraining the whole model, LoRA lets us adjust only a small part of it, which saves time and resources.

At first it was a little confusing to understand how it works, but once we got it running, it made sense why this method is used a lot. It's much faster than full fine-tuning but still gives good results.

System Integration Description

We connected the adapted model into our existing system using FastAPI and Streamlit.

The process works like this:
1. The user enters a question in the Streamlit app

2. The request goes to the FastAPI backend
3. The fine-tuned model generates a response
4. The answer is sent back and shown to the user

This helped us see how everything fits together in a real pipeline instead of just running the model separately.

Evaluation Results

We tested the system using different questions before and after applying domain adaptation.

After adaptation, the answers were:
- More structured
- More focused on the topic
- Less generic

The difference wasn't huge in every case, but overall the responses were definitely better than before.

Impact on Project Performance

The main improvement we saw was in how consistent the answers were. The model stayed more within the domain and didn't drift as much.

Even though the dataset wasn't very large, using LoRA still made a noticeable difference. It showed that you don't always need a massive dataset to improve a system.

Contribution Table

Ibrahim Alborno – Dataset creation, LoRA setup, backend integration, evaluation (50%)
Immanuel Olaoye – Streamlit UI, testing, prompt improvements, organization (50%)